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Financial Analytics Lab

GROUP 11

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01 Answers

[Link to PYTHON CODE](#)

Q1.

After running our code as mentioned in the link above we have the following results pertaining to the unit root and stationarity inferences for all the variables present in the data provided:

Variable	Optimum Lag	P-Value	Stationarity
Inflation	1	0.0550	Likely Non-stationary
Unemployment Rates	2	0.3266	Likely Non-stationary
Repo Rate	7	0.1404	Likely Non-stationary
Reverse Repo Rate	7	0.1377	Likely Non-stationary

Inference:

- The p-values for "Unemployment Rates", "Repo Rate", "Reverse Repo Rate" are significantly greater than the threshold value of 0.05 indicating strong non-stationarity.
- For "Inflation" a p-value of 0.0550 indicates weak non-stationarity relative to other variables.
- Since all the variables are Non-Stationary a unit root does exist for all the variables at an optimum lag (minimum AIC):
 - Inflation : unit root exists at OPTIMUM LAG = 1
 - Unemployment Rates: unit root exists at OPTIMUM LAG = 2
 - Repo Rate : unit root exists at OPTIMUM LAG = 7
 - Reverse Repo Rate : unit root exists at OPTIMUM LAG = 7

Answers

Q2.

Causality Inference as given from code:

```
Granger Causality Test for unemployment Rates with Inflation:  
Lag 1: p-value = 6.281744785558605e-08  
Lag 2: p-value = 0.00016368980404629397  
Lag 3: p-value = 0.00044386338548888947  
Lag 4: p-value = 0.00031235584542798245  
Lag 5: p-value = 0.0004144419348647724  
Lag 6: p-value = 0.0009732386712970456  
Lag 7: p-value = 0.0015049203909040002
```

Forecasted Values for next 5 years (Inflation):

```
Forecast for Inflation for the next 5 years (2024-2030) using unemployment Rates:  
2024Q1    3.972854  
2024Q2    4.047742  
2024Q3    4.165006  
2024Q4    4.305630  
2025Q1    4.456305  
2025Q2    4.607949  
2025Q3    4.754586  
2025Q4    4.892492  
2026Q1    5.019556  
2026Q2    5.134798  
2026Q3    5.238020  
2026Q4    5.329537  
2027Q1    5.409993  
2027Q2    5.480218  
2027Q3    5.541139  
2027Q4    5.593705  
2028Q1    5.638850  
2028Q2    5.677461  
2028Q3    5.710360  
2028Q4    5.738299
```

Answers

Q4. Causality Inference as given from code(Unemployment)

```
Granger Causality Test for unemployment Rates with Inflation:
Lag 1: p-value = 6.281744785558605e-08
Lag 2: p-value = 0.00016368980404629397
Lag 3: p-value = 0.00044386338548888947
Lag 4: p-value = 0.00031235584542798245
Lag 5: p-value = 0.0004144419348647724
Lag 6: p-value = 0.0009732386712970456
Lag 7: p-value = 0.0015049203909040002

Model summary for unemployment Rates:
SARIMAX Results
=====
Dep. Variable:          Inflation      No. Observations:      164
Model:                ARIMA(1, 1, 1)  Log Likelihood         -6.780
Date:                 Thu, 21 Mar 2024  AIC                     19.561
Time:                  11:15:14         BIC                     28.842
Sample:                0               HQIC                    23.329
                                - 164
Covariance Type:       robust
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
ar.L1          0.5241     0.161      3.247     0.001     0.208     0.841
ma.L1          0.1771     0.168      1.053     0.292    -0.153     0.507
sigma2         0.0634     0.011     5.877     0.000     0.042     0.085
=====
Ljung-Box (L1) (Q):                0.00  Jarque-Bera (JB):                66.00
Prob(Q):                           0.97  Prob(JB):                  0.00
Heteroskedasticity (H):             0.55  Skew:                        0.71
Prob(H) (two-sided):                0.03  Kurtosis:                     5.78
=====
```

Forecasted Values for next 6 years (Unemployment):

```
Lag length: 1

Forecast for Inflation for the next 6 years (2024-2030) using unemployment Rates:
2024Q1 3.888002
2024Q2 3.846595
2024Q3 3.824892
2024Q4 3.813517
2025Q1 3.807554
2025Q2 3.804429
2025Q3 3.802791
2025Q4 3.801933
2026Q1 3.801483
2026Q2 3.801247
2026Q3 3.801123
2026Q4 3.801058
2027Q1 3.801024
2027Q2 3.801006
2027Q3 3.800997
2027Q4 3.800992
2028Q1 3.800990
2028Q2 3.800988
2028Q3 3.800988
2028Q4 3.800987
2029Q1 3.800987
2029Q2 3.800987
2029Q3 3.800987
2029Q4 3.800987
```

Answers

Q4. Causality Inference as given from code(Repo Rate)

```

Granger Causality Test for Repo Rate with Inflation:
Lag 1: p-value = 4.73996906050521e-08
Lag 2: p-value = 1.246432043854131e-06
Lag 3: p-value = 3.6004662753885245e-06
Lag 4: p-value = 2.0928651135494786e-06
Lag 5: p-value = 3.4309660747565476e-06
Lag 6: p-value = 5.999488024956886e-06
Lag 7: p-value = 4.85460144938726e-06

Model summary for Repo Rate:

=====
SARIMAX Results
=====
Dep. Variable:          Inflation    No. Observations:          164
Model:                ARIMA(1, 1, 1)  Log Likelihood             -6.780
Date:                 Thu, 21 Mar 2024  AIC                          19.561
Time:                 11:15:15         BIC                         28.842
Sample:               0               HQIC                        23.329
                             - 164
Covariance Type:      robust

=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
ar.L1         0.5241     0.161      3.247     0.001     0.208     0.841
ma.L1         0.1771     0.168      1.053     0.292    -0.153     0.507
sigma2        0.0634     0.011     5.877     0.000     0.042     0.085
=====
Ljung-Box (L1) (Q):                0.00  Jarque-Bera (JB):                66.00
Prob(Q):                           0.97  Prob(JB):                  0.00
Heteroskedasticity (H):             0.55  Skew:                        0.71
Prob(H) (two-sided):               0.03  Kurtosis:                    5.78
=====

```

Forecasted Values for next 6 years (Repo Rate):

```

Forecast for Inflation for the next 6 years (2024-2030) using Repo Rate:
2024Q1    3.888002
2024Q2    3.846595
2024Q3    3.824892
2024Q4    3.813517
2025Q1    3.807554
2025Q2    3.804429
2025Q3    3.802791
2025Q4    3.801933
2026Q1    3.801483
2026Q2    3.801247
2026Q3    3.801123
2026Q4    3.801058
2027Q1    3.801024
2027Q2    3.801006
2027Q3    3.800997
2027Q4    3.800992
2028Q1    3.800990
2028Q2    3.800988
2028Q3    3.800988
2028Q4    3.800987
2029Q1    3.800987
2029Q2    3.800987
2029Q3    3.800987
2029Q4    3.800987
Forecast: 0 DEC      Name: predicted mean      dtype: float64

```

Answers

Q4. Causality Inference as given from code(Reverse Repo Rate)

```

Granger Causality Test for Reverse Repo Rate with Inflation:
Lag 1: p-value = 4.272054780646611e-08
Lag 2: p-value = 1.1687186852490635e-06
Lag 3: p-value = 3.5167086265869603e-06
Lag 4: p-value = 1.957595198974856e-06
Lag 5: p-value = 3.429330742477378e-06
Lag 6: p-value = 5.9181152029151946e-06
Lag 7: p-value = 4.595134270142103e-06

Model summary for Reverse Repo Rate:
SARIMAX Results
=====
Dep. Variable:          Inflation      No. Observations:          164
Model:                ARIMA(1, 1, 1)  Log Likelihood            -6.780
Date:                  Thu, 21 Mar 2024  AIC                        19.561
Time:                  11:15:15         BIC                       28.842
Sample:                0               HQIC                      23.329
Covariance Type:       robust
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ar.L1          0.5241      0.161      3.247      0.001      0.208      0.841
ma.L1          0.1771      0.168      1.053      0.292     -0.153      0.507
sigma2         0.0634      0.011      5.877      0.000      0.042      0.085
=====
Ljung-Box (L1) (Q):          0.00  Jarque-Bera (JB):          66.00
Prob(Q):                    0.97  Prob(JB):              0.00
Heteroskedasticity (H):      0.55  Skew:                0.71
Prob(H) (two-sided):         0.03  Kurtosis:            5.78
=====

```

Forecasted Values for next 6 years (Reverse Repo Rate):

```

Forecast for Inflation for the next 6 years (2024-2030) using Reverse Repo Rate:
2024Q1    3.888002
2024Q2    3.846595
2024Q3    3.824892
2024Q4    3.813517
2025Q1    3.807554
2025Q2    3.804429
2025Q3    3.802791
2025Q4    3.801933
2026Q1    3.801483
2026Q2    3.801247
2026Q3    3.801123
2026Q4    3.801058
2027Q1    3.801024
2027Q2    3.801006
2027Q3    3.800997
2027Q4    3.800992
2028Q1    3.800990
2028Q2    3.800988
2028Q3    3.800988
2028Q4    3.800987
2029Q1    3.800987
2029Q2    3.800987
2029Q3    3.800987
2029Q4    3.800987
Freq: Q-DEC, Name: predicted_mean, dtype: float64

```